

ASTRODYNAMICS I: HOMEWORK #2

Due: Mon, Apr 16, 11:00AM, 2018

For the following problems, use $\mu=398,600.4418\text{km}^3/\text{sec}^2$ and $R_E=6,378\text{km}$ if necessary.

1. Calculate the eccentric anomaly and true anomaly for $M = 215.4^\circ$ and $e = 0.3$.
2. A rogue asteroid has just been discovered. Your job is to determine how much time we have to intercept and destroy the object, if $a = -2,797.425\,069\text{km}$, $e = 2.8$, $i = 23^\circ$, $\nu = 249.27^\circ$, and $r = 2,150,138.804\,411\text{km}$.
3. You have received two position vectors for a satellite. How much time has elapsed between these two positions? ($p = 6,681.571\text{km}$)

$$\vec{r}_0 = -2,574.9533\hat{i} + 4,267.0671\hat{j} + 4,431.5026\hat{k} \text{ km}$$

$$\vec{r} = 2,700.6738\hat{i} - 4,303.5378\hat{j} - 4,358.2499\hat{k} \text{ km}$$

4. Rader readings determine that an object is located at

$$r = 1.8\hat{K} \text{ DU} \text{ and } v = (0.8\hat{i} + 0.6\hat{K})\text{DU/TU}$$

Determine $p, e, I, \Omega, \omega, \nu, u, \lambda_{true}$.

5. For the following orbital elements:

$$p = 0.42\text{DU}, e = 0.60, I = 90^\circ, u = 90^\circ, \Omega = 180^\circ, \omega = 260^\circ, \nu = 190^\circ, \lambda_{true} = 270^\circ$$

- a. Express $\vec{r}$ and $\vec{v}$ for the satellite in the perifocal coordinate system along the unit vectors $\hat{P}, \hat{Q}, \hat{W}$.
- b. By a suitable coordinate transformation technique, express $\vec{r}$ and $\vec{v}$ in the geocentric equatorial system in the $\hat{i}, \hat{j}, \hat{K}$ axis.